AYMAN ELBERY

Senior Full-Stack Engineer / Technical Team Leader / Software Engineer

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SUMMARY

Senior Full-Stack Software Engineer with 10+ years of experience designing and delivering scalable, enterprise-grade applications and digital platforms. Expert in Angular, NestJS, Node.js, Laravel, TypeScript, and modern microservices architectures. Strong experience in enterprise authentication and SSO using Keycloak, including secure BFF and OAuth/OIDC architectures. Skilled in RESTful APIs, event-driven systems, RabbitMQ, Redis, Docker, Linux, and SQL/NoSQL databases. Experienced in complex enterprise and government integrations, high-availability deployments, and restricted/offline environments. Hands-on experience building AI-powered platforms using LLMs and RAG architectures. Proven ability to modernize legacy systems, establish engineering standards, and solve complex production challenges. Combines hands-on development expertise with software architecture, technical leadership, mentoring, and a strong focus on security, scalability, performance, and maintainability.

ACHIEVEMENTS

- Designed and implemented secure **SSO and Identity & Access Management architectures using Keycloak**, supporting multiple enterprise systems and centralized authentication.
- Successfully integrated enterprise applications with **national and government services**, including **Nafath and Yakeen**, enabling secure digital identity verification.
- Designed **microservices and event-driven architectures** using NestJS, Node.js, RabbitMQ, Redis, and API-based integrations for complex enterprise workflows.
- Built **AI-powered RAG and document intelligence solutions**, enabling users to interact with organizational knowledge and documents using LLMs.
- Modernized legacy applications and integration of workflows into **scalable, maintainable architectures**, improving system reliability and development efficiency.

EXPERIENCE

The Royal Saudi Air Force (RSAF) **05/2023 - Present**
Senior Full-Stack Engineer Riyadh, Saudi Arabia

- Lead **frontend architecture (Angular)**, establishing reusable components, shared libraries, coding standards, and scalable application structures in an offline environment.
- Design and implement **RESTful APIs, microservices, BFF architectures, and event-driven solutions** for complex enterprise workflows and system integrations.
- Architect and maintain centralized **Identity & Access Management and SSO solutions using Keycloak and Oracle IDM**, including OAuth 2.0/OIDC, RBAC, MFA/OTP, and secure authentication flows.

- Integrate enterprise platforms with **national services such as NIC travel permission service**, as well as internal and third-party enterprise systems.
- Develop and integrate **AI-powered solutions**, including RAG-based document intelligence platforms and LLM-enabled enterprise applications.
- Deploy and troubleshoot applications across **Linux, Docker, Apache, load-balanced, high-availability, and restricted/offline environments**.
- Provide **technical leadership and mentoring**, conduct architecture and code reviews, resolve complex production issues, and promote security, performance, maintainability, and engineering best practices.

Saudi Electronic University (SEU) **11/2019 - 05/2023**
Senior Full-Stack Software Engineer Riyadh, Saudi Arabia

- Designed, developed, and maintained **large-scale university platforms serving 70,000+ users**, ensuring scalability, reliability, and high availability.
- Developed modern web applications using **Angular, Laravel, PHP, Node.js, TypeScript, and RESTful APIs** across multiple academic and administrative systems.
- Designed and implemented **enterprise APIs and system integrations** connecting university platforms with internal, external, and government services.
- Implemented secure **authentication, authorization, SSO using Oracle IDM, and role-based access control** for applications serving students, faculty, and administrative users.
- Developed complex academic and administrative workflows involving **student services, learning platforms, reporting, and digital transformation initiatives**.
- Integrated frontend and backend systems with **Oracle**, **SQL Server, MySQL, and other enterprise data sources**, optimizing queries and application performance.
- Integrate enterprise platforms with **national services such as Nafath and Yakeen**, as well as internal and third-party enterprise systems.
- Deploy and troubleshoot applications across **Linux, IIS, Docker, Apache, load-balanced, high-availability, and restricted/offline environments**.

Buseet **01/2019 - 11/2019**
Senior Software Engineer Cairo, Egypt

- Developed and maintained **transportation and mobility platforms** supporting daily operational and customer-facing services.
- Developed backend services and APIs supporting **trip management, bookings, users, routes, vehicles, and transportation operations**.
- Built administrative dashboards and operational tools for **monitoring trips, managing transportation services, and supporting business operations**.
- Diagnosed and resolved **production issues, performance bottlenecks, and application defects**, improving platform stability and reliability.
- Implemented **real-time geospatial tracking and location-based services using Tile38**, including vehicle location tracking, geofencing, and proximity-based queries to support transportation and fleet operations.

Afaqy **12/2017 - 01/2019**
Full-Stack Engineer Cairo, Egypt

- Developed and maintained **GPS tracking and fleet management platforms** for real-time monitoring of vehicles, assets, and transportation operations.
- Built scalable web applications and backend services using **Angular, PHP/Laravel, JavaScript/TypeScript, and RESTful APIs**.
- Implemented **real-time vehicle tracking and location visualization**, processing GPS coordinates and telemetry data received from tracking devices.
- Built interactive **map-based dashboards** to visualize vehicle locations, movements, routes, and operational information.
- Collaborated with engineering and product teams to transform **fleet-management and location-based business requirements into production-ready solutions**.

Speakol **01/2016 - 08/2016**
Backend Engineer Cairo, Egypt

- Built and enhanced backend services using **PHP, Laravel, MySQL, JavaScript, and RESTful APIs**.
- Developed features for **content recommendation, publisher management, advertising campaigns, and user engagement**.
- Built and integrated **RESTful APIs** connecting internal platform components, publisher websites, and third-party services.
- Optimized **database queries and backend processes** to improve application response times and overall platform performance.
- Developed administrative dashboards and internal tools for managing **publishers, content, campaigns, and platform operations**.
- Refactored and modernized **legacy PHP applications built with the Phalcon Framework**, improving code quality, maintainability, performance, and overall application architecture while preserving existing business functionality.

EDUCATION

Bachelor's Degree in Computer Science

Faculty of Computers & Information, Benha University | Egypt

Graduation Year: 2016 | **Overall Grade:** Good

- Strong academic foundation in **Software Engineering, Data Structures, Algorithms, Database Systems, Object-Oriented Programming, Computer Networks, and Operating Systems**.
- Developed practical software projects applying **programming, database design, problem-solving, and software development principles**.

CERTIFICATIONS

- AWS Technical Essentials **(AWS)**
- Docker & Kubernetes: The Practical Guide Certificate **(Udemy)**

SKILLS

- **Frontend:** Angular (v9–v20), TypeScript, JavaScript, RxJS, NgRx, Angular Material, Tailwind CSS, HTML5, CSS3, Responsive Design, Bootstrap.
- **Backend & APIs:** NestJS, Node.js, Laravel, PHP, Express.js, Phalcon, RESTful APIs, SOAP, Microservices, BFF Architecture.
- **Identity & Security:** Keycloak, SSO, OAuth 2.0, OpenID Connect (OIDC), PKCE, JWT, Oracle IDM.
- **Databases & Caching:** Oracle, SQL Server, PostgreSQL, MySQL, MongoDB, Redis, TypeORM.
- **Messaging & Real-Time:** RabbitMQ, WebSockets, Event-Driven Architecture, Asynchronous Processing.
- **Maps & Geospatial:** Tile38, Leaflet, Geofencing, GPS Tracking, Location-Based Services, Geospatial Queries.
- **AI & LLM:** RAG Architecture, LLM Integration, Vector Databases, Qdrant, AI-Powered Document Search, Prompt Engineering.
- **DevOps & Infrastructure:** Docker, Docker Compose, Linux, Red Hat Enterprise Linux, Apache HTTPD, Nginx, Load Balancing, Systemd, Offline/Air-Gapped Deployments.
- **Enterprise Integration:** Software AG webMethods, REST/SOAP Integrations, Nafath, Yakeen, NIC Government & Third-Party APIs & SMS and payment gateways.
- **Architecture & Engineering:** Clean Architecture, SOLID Principles, Design Patterns, System Design, API Design, High Availability, Scalability, Performance Optimization.
- **Tools:** Git, GitHub, GitLab, VS Code, Postman, Swagger/OpenAPI, Jira, npm, Composer, Azure.

LANGUAGES

- **Arabic:** Native
- **English:** Professional Working Proficiency